Charles Ciampa

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Education

Northeastern University; Khoury College of Computer Sciences — Boston, MA
M.S. in Artificial Intelligence, Concentration in Machine Learning; GPA: 4.0 — Expected May 2027
B.S. in Computer Science and Physics, Minor in Mathematics; GPA: 3.3 — August 2025

Related Courses: Reinforcement Learning, Natural Language Processing, Deep Learning, Computer Vision, Electronic, Astrophysics Machine Learning & Data Mining, Robotic Science & Systems, Robotic Sensing & Navigation, Statistics & Stochastic Processes

Skills

Computer Languages: Python | Swift | Java | JavaScript | Rust | Typescript | Latex | HTML | CSS | Racket | C++ | R
Software: Git/GitHub | NumPy & Pandas | PyTorch | ROS2 | OpenCV | React | Jira | Selenium | Vue | SPSS

Experience

S1 Industries PBC Inc. — August 2024 – December 2024
Engineering Intern — Medford, MA

- Performed 10+ explorations of technologies and problems using methodology developed by MIT professor Luis Perez-Breva for his startups focused iTeams class, including reading research for each new technology and proposing potential companies to create
- Spearheaded the development of an Augmented Reality Tic-Tac-Toe game with options for multiplayer remote playing for VisionOS's mixed immersive environment using Swift, and support for Apple Personas over SharePlay
- Created app specifications, testing procedures, and utilized TestFlight for internal testing with a small community of users to ensure the Tic-Tac-Toe app met requirements
- Launched an internal AI project using local LLMs to read technical documents and access the internet through a local instance of SearXNG and mapped over 70 separate AI and LLM tools and software with 3 views, including relevant logos, summaries, use cases, rankings, and applicable licenses related to usage or development

Gentuity — July 2023 – December 2023
Medical Device Testing Engineer Associate — Sudbury, MA

- Engineered a well-documented Python program for point tracking in high-speed camera footage, offering users flexibility with customizable algorithms and seamless data export in multiple formats
- Developed a user-friendly interface and adaptable tracking settings, providing accessibility through Python scripts, command line interface, options for selection of specific video frame areas and time intervals for efficient analysis, and a graphical user interface (GUI), optimizing user interaction and usability
- Conducted rigorous testing on both software and catheter integration with the OCT device, ensuring robust verification and validation and meticulous data collection for quality assurance purposes
- Executed tests and collected data to secure FDA, BSI, and PMDA submissions for a new catheter

AI Club — May 2023 – April 2025
Founding Member and Vice President — Boston, MA

- Pioneered a project proposal, assignment, and mentoring system for club members to collaborate in teams and raise funding, as well as aiding in the development of the current project (SwarmScape) to monitor greenhouses with swarms of robots
- Hosted an AI hackathon, in collaboration with the Babson AI Club, with over 80 participants and 11 expert speakers and judges from industry and academia
- Managed an executive board of 8 people to coordinate weekly meetings, schedule events for students, organize lab tours, and ideate for future semesters

Projects

Chess Bot Player Move Predictor — March 2025 – April 2025

- Developed a neural network combining CNN, LSTM, and ResNet architectures inspired by Google DeepMind's AlphaZero to predict player moves, training on 500K games from 36.5 million chess positions using PyTorch
- Preprocessed 100M games from Lichess PGN files using Rust, extracting and formatting data into H5 format in under 30 minutes
- Worked with team of 4 comparing AI chess methodologies, compiling results into AAAI-formatted paper *AI Models For Chess*

Robotic Photographer for 3D Gaussian Splatting — March 2025 – April 2025

- Deployed ROS Noetic on LoCoBot WX250 6 DOF on Create 3 mobile base to autonomously detect, approach, and circumnavigate objects using object segmentation without April tags, capturing RGBD imagery for 3D Gaussian Splatting reconstruction
- Implemented ZeroMQ-based networking to stream camera data to remote workstation, collaborating with team of 4 to achieve sub-minute capture workflow documented in paper *Robotic Photographer for 3D Gaussian Splatting*

Interests

Languages: English | Spanish (Fluent and Literate)
Other: Biking | Rowing | Traveling | Astronomy | Open-World Video Games (Minecraft, Red Dead 2) | Reading Manga